

ICS 604

Applied Data Science

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Reminder

- Homework Assignment 2
 - Due: **11:59 PM** on **Friday, March 13**
 - Please carefully read and follow the assignment instructions and problem requirements.
 - Submit only the Jupyter Notebook file (.ipynb) to Lamaku.
 - Do not upload the data file.
 - Make sure to **rename your notebook file** as instructed.
- No classes next week: Spring recess

Project proposal

- Due: **11:59 PM, Monday, March 23**
 - Create a **PDF file** and submit it to Lamaku.
 - Maximum 2 pages (~ 700 – 900 words)
- The proposal **must include a title**.
- You may include **as many figures as needed**; they do not count toward the page limit.
- The proposal will be evaluated based on **how well it addresses the items in the project requirements document**, not on the idea or scope itself.

Lecture 15

- Hypothesis Testing
 - Comparing means
 - The p-value

Previously... Hypothesis testing

- A statistical method for making inferences about populations from sample data.
- Two main hypothesis:
 - **Null hypothesis (H_0):** Assumes no change, difference, or effect — any observed result is due to random chance. Serves as the baseline for testing.
 - **Alternative hypothesis (H_a):** Opposes the null hypothesis, indicating a real change, difference, or effect beyond random chance.
- The goal is to use sample data to determine whether there is sufficient statistical evidence to reject H_0 or fail to reject H_0 in favor of H_a .

Example: Data from a theoretical distribution

- Suppose the data come from two normal distributions:
 - Means: $\mu_1 = 232$ and $\mu_2 = 219$
 - Standard deviations: $\sigma = 12$ (same for both distributions)
- For example, this could represent the average ad time in minutes viewed daily on YouTube versus Vimeo.
 - We have 90 days worth of viewing data.

```

np.random.seed(42)

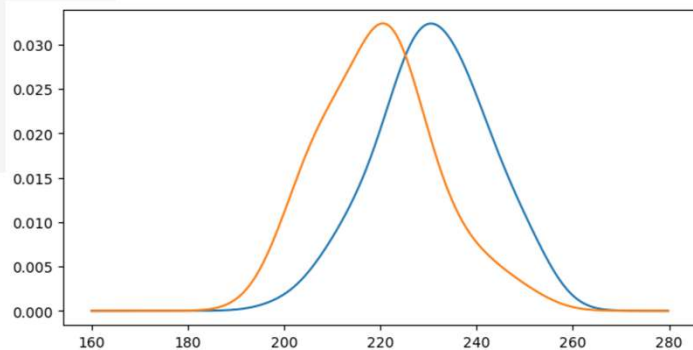
youtube = np.random.normal(232, 12, 90)
vimeo = np.random.normal(219, 12, 90)

kde_1 = sp.stats.gaussian_kde(youtube, bw_method=0.5)
kde_2 = sp.stats.gaussian_kde(vimeo, bw_method=0.5)

# Estimate kernel density for the support (x-axis)
x_values = np.arange(160, 280, 0.25)
x_densities_1 = kde_1.evaluate(x_values)
x_densities_2 = kde_2.evaluate(x_values)

plt.figure(figsize=(8, 4))
plt.plot(x_values, x_densities_1)
plt.plot(x_values, x_densities_2)
plt.show()

```



Assessing the variability of our statistic: Different samples

- The difference between the means is a useful statistic to assess the difference between both samples.

- We call a statistic computed on the observed data the *observed statistic*.

```
np.mean(youtube) - np.mean(vimeo)
```

11.159994107248792

- Given that the data consists of random samples, collecting new data will undoubtedly lead to a change in observed statistic.
- We need a distribution of the difference between the two samples under the null hypothesis to compare our observed statistic.
 - This is the *sampling distribution of the test statistic*.

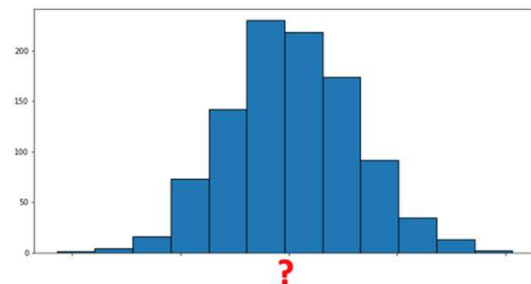
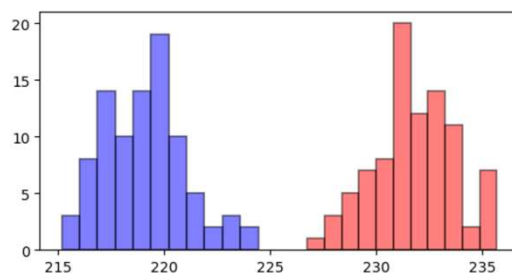
Assessing the variability of our statistic: Different samples (cont'd.)

- We will start with an example where the samples are clearly (visually) different.
 - To do so, we generate a dataset with $\sigma = 2$.

```
youtube = np.random.normal(232, 2, 90)  
vimeo = np.random.normal(219, 2, 90)
```

Question

- What is the mean of the distribution of the differences between the means?



Sampling distribution of the differences in means

```
means_difference = []
```

```
for _ in range(1000):
```

```
    youtube = np.random.normal(232, 2, 90)
```

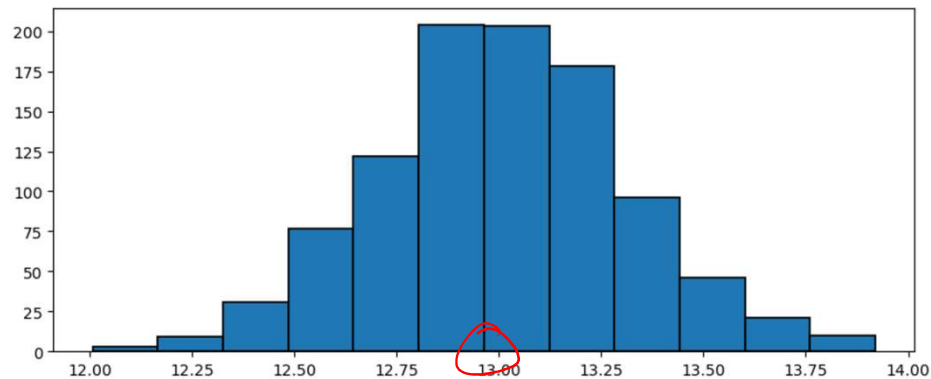
```
    vimeo = np.random.normal(219, 2, 90)
```

```
    means_difference.append(np.mean(youtube) - np.mean(vimeo))
```

```
plt.figure(figsize=(8, 4))
```

```
plt.hist(means_difference, edgecolor='k', linewidth=1.2, bins=12)
```

```
plt.show()
```



} simulate sampling from population

What if samples are similar?

- What does the difference in means between two samples of the same distribution look like on average?

```
youtube_1 = np.random.normal(232, 2, 90)
```

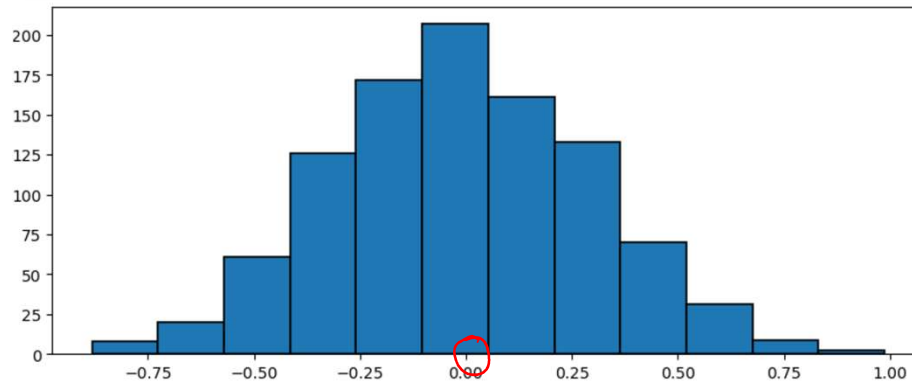
```
youtube_2 = np.random.normal(232, 2, 90)
```

```
means_difference.append(np.mean(youtube_1) - np.mean(youtube_2))
```

```
means_difference = []

for _ in range(1000):
    youtube_1 = np.random.normal(232, 2, 90)
    youtube_2 = np.random.normal(232, 2, 90)
    means_difference.append(np.mean(youtube_1) - np.mean(youtube_2))

plt.figure(figsize=(8, 4))
plt.hist(means_difference, edgecolor='k', linewidth=1.2, bins=12)
plt.show()
```



What if samples are similar (cont'd.)

- Data in youtube_1 and youtube_2 come from the same distribution.
- Instead of generating two datasets separately, we can:
 1. Generate the data in the same call to `np.random.normal` and then split the data into youtube_1 and youtube_2.
 2. We could even permute the data and split it randomly across youtube_1 and youtube_2.

```

# Generating the data in the same call,
# shuffling and splitting the data does not change the results

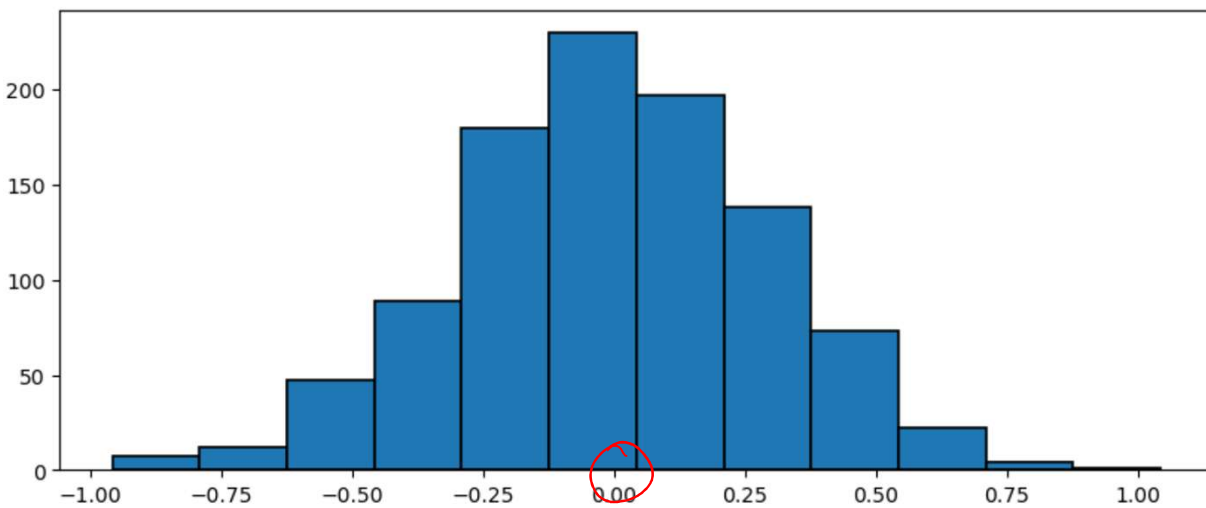
means_difference = []

for _ in range(1000):
    # Generate 180 days worth of data in each iteration
    all_data = np.random.normal(232, 2, 180)

    # shuffle the data and then split it over youtube_1 and youtube_2
    np.random.shuffle(all_data)
    youtube_1 = all_data[0:90]
    youtube_2 = all_data[90:]
    means_difference.append(np.mean(youtube_1) - np.mean(youtube_2))

plt.figure(figsize=(8, 4))
plt.hist(means_difference, edgecolor='k', linewidth=1.2, bins=12)
plt.show()

```

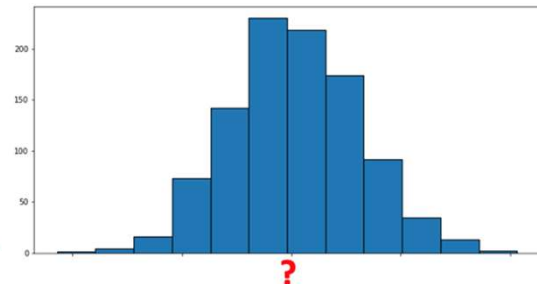


Question 1

- What would the distribution of differences between the means look like if we concatenate (merge) and then permute (shuffle) the Vimeo and YouTube data?

```
all_data = np.concatenate([youtube, vimeo])

for _ in range(1000):
    np.random.shuffle(all_data)
    S1 = all_data[:90]
    S2 = all_data[90:]
    means_difference.append(np.mean(S1) - np.mean(S2))
```



Sampling distribution of the differences in means under the null hypothesis

Concatenated and permuted YouTube and Vimeo samples



```
means_difference = []
```

```
youtube = np.random.normal(232, 2, 90)
```

```
vimeo = np.random.normal(219, 2, 90)
```

```
all_data = np.concatenate([youtube, vimeo])
```

```
for _ in range(1000):
```

```
    np.random.shuffle(all_data)
```

```
    youtube_s = all_data[:90]
```

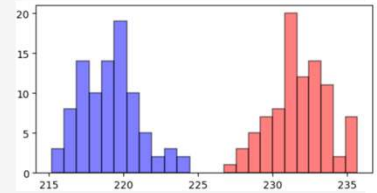
```
    vimeo_s = all_data[90:]
```

```
    means_difference.append(np.mean(youtube_s) - np.mean(vimeo_s))
```

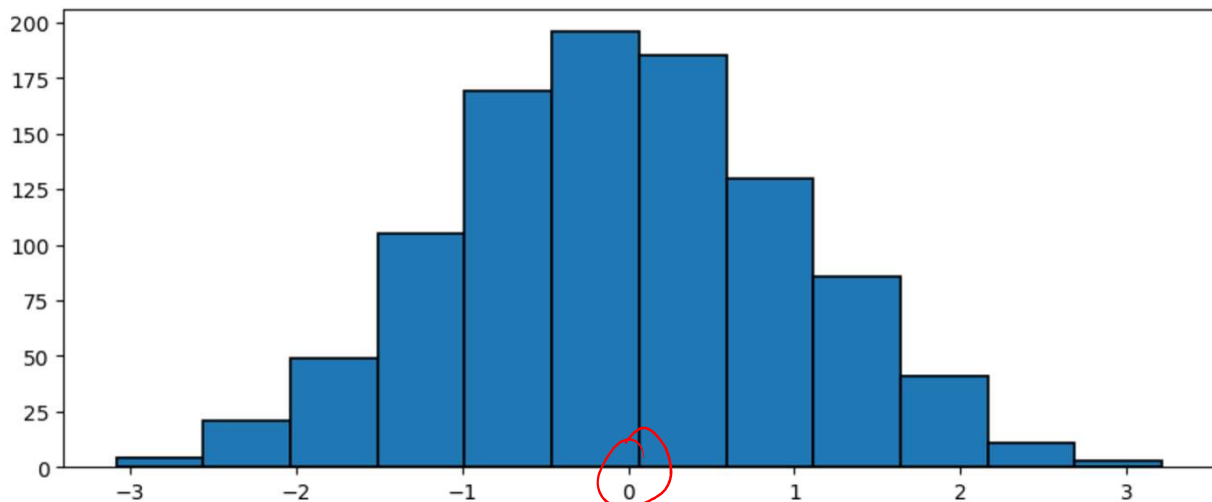
```
plt.figure(figsize=(8, 4))
```

```
plt.hist(means_difference, edgecolor='k', linewidth=1.2, bins=12)
```

```
plt.show()
```



large pop.



Questions

Question 1: What would the distribution of differences between the means look like if we concatenate (merge) and then permutate (shuffle) the Vimeo and YouTube data?

close to

Question 2: What would the distribution of the shuffled difference between Vimeo and YouTube look like if the samples are not balanced?

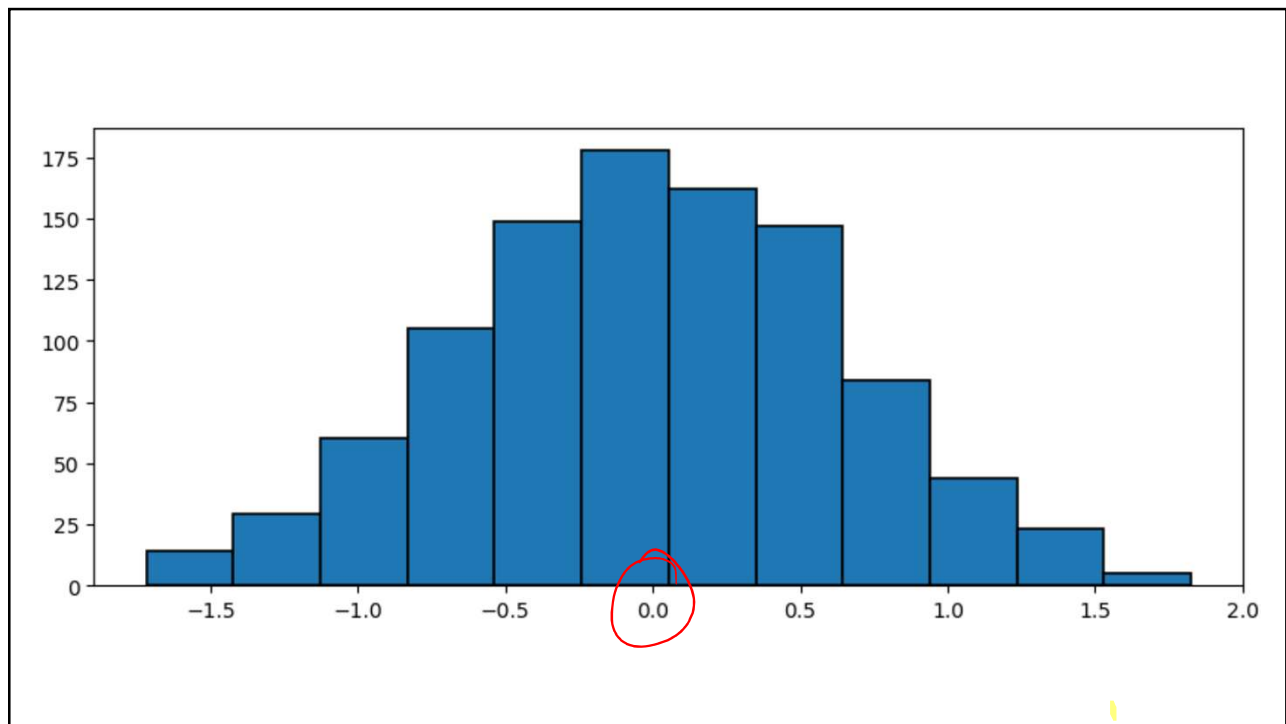
- For example, one sample has 90 individuals and the other has four times as many.

```
means_difference = []

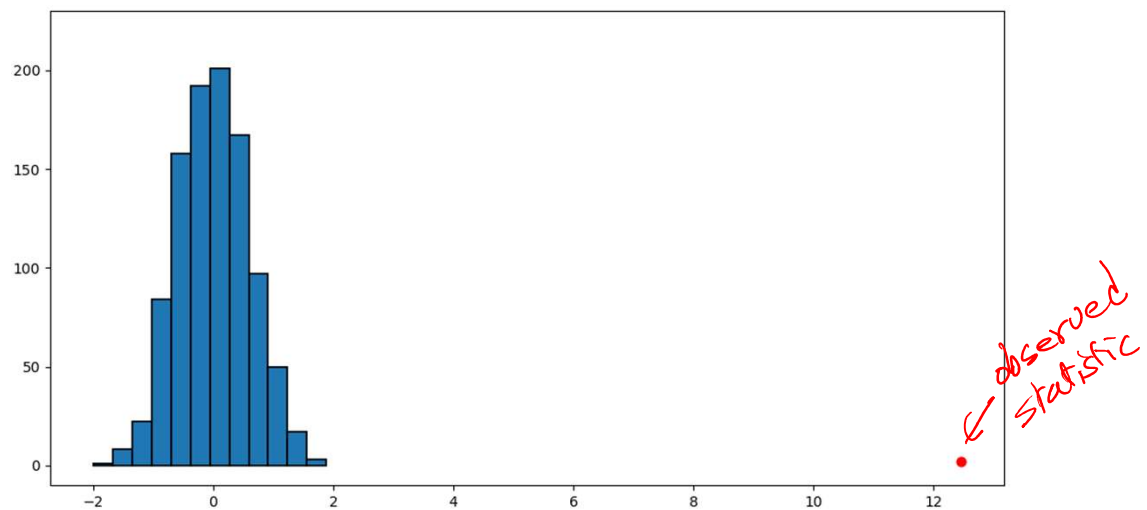
youtube = np.random.normal(232, 2, 90)
vimeo = np.random.normal(219, 2, 360)
all_data = np.concatenate([youtube, vimeo])

for _ in range(1000):
    np.random.shuffle(all_data)
    youtube_s = all_data[:90]
    vimeo_s = all_data[90:]
    means_difference.append(np.mean(youtube_s) - np.mean(vimeo_s))

plt.figure(figsize=(8, 4))
plt.hist(means_difference, edgecolor='k', linewidth=1.2, bins=12)
plt.show()
```



Original observation vs. distribution under the null hypothesis



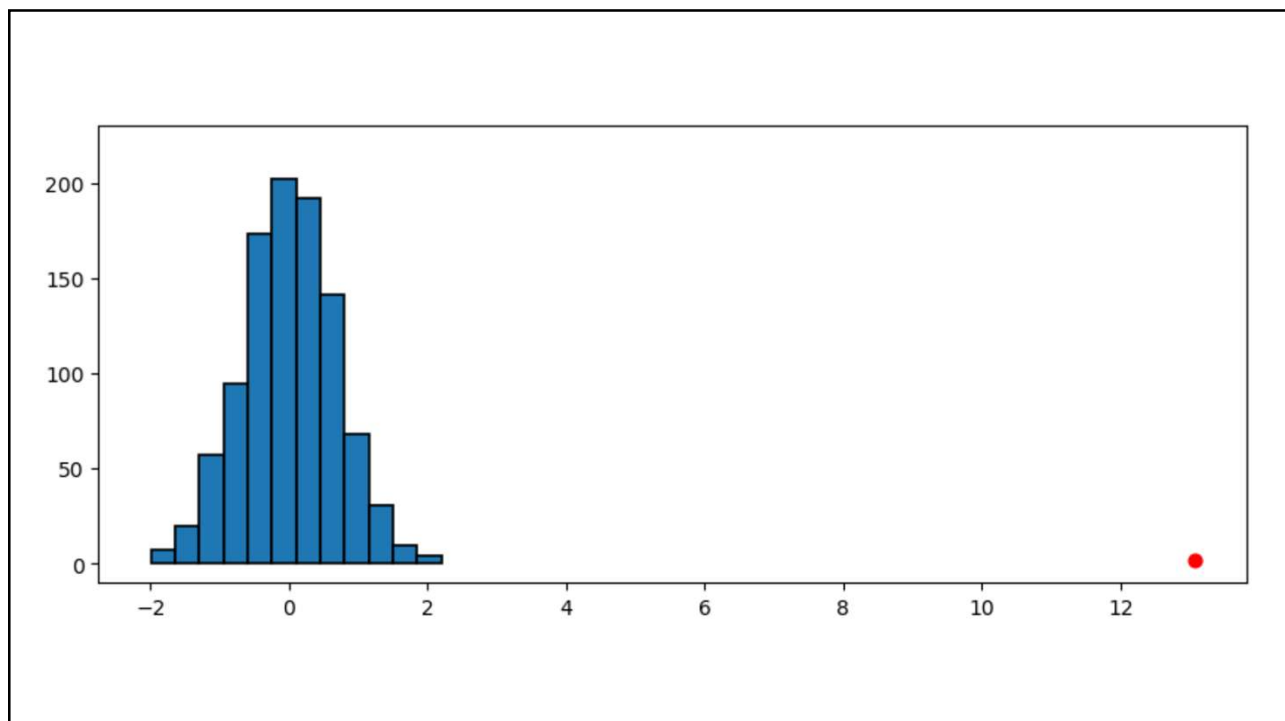
```
means_difference = []

youtube = np.random.normal(232, 2, 90)
vimeo = np.random.normal(219, 2, 360)
all_data = np.concatenate([youtube, vimeo])

Building null dist
for _ in range(1000):
    np.random.shuffle(all_data)
    youtube_s = all_data[:90]
    vimeo_s = all_data[90:]
    means_difference.append(np.mean(youtube_s) - np.mean(vimeo_s))

plt.figure(figsize=(8, 4))
plt.hist(means_difference, edgecolor='k', linewidth=1.2, bins=12)
plt.scatter(np.mean(youtube) - np.mean(vimeo), 2, color="r", s=40)
plt.ylim(-10, 230)
plt.show()
```

↓ observed statistic



What have we learned?

- The difference between the means of samples from the same distribution has a mean close to zero.
- This is very useful for testing the null hypothesis.
 - Under the null hypothesis, we assume that the difference between the values observed across two samples is attributed to chance alone.
- Comparing the observed difference between the means of two samples to that obtained if samples are from the same distribution shows how similar/dissimilar the samples while accounting for sampling variability.

Comparing samples from similar distributions

- What does a plausible observed difference of means look like?
- Example: Is there a difference between the spending of males and females on your e-commerce platform?
 - Again, we will generate the data from two distributions.
 - The distributions are different but close:
 - $\mu_{male} = 466$ and $\mu_{female} = 450$
 - The standard deviation is the same for both populations but relatively large ($\sigma = 89$).

```
np.random.seed(129)
```

```
male_spending = np.random.normal(466, 89, 230)
```

```
female_spending = np.random.normal(450, 89, 190)
```

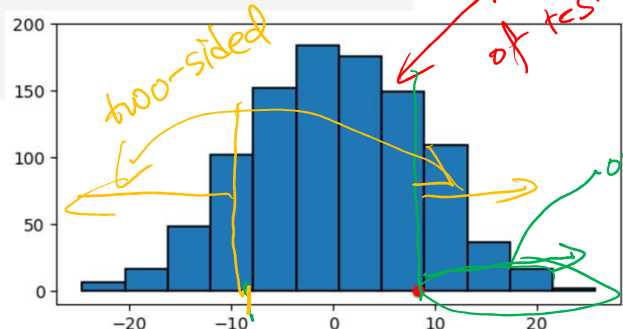
```
diff_of_means = np.mean(male_spending) - np.mean(female_spending)  
print(f"Difference between means is {diff_of_means}")
```

Difference between means is 8.356736308940526

↖ observed
statistic

```
all_data = np.concatenate([male_spending, female_spending])  
means_difference = []  
  
for _ in range(1000):  
    np.random.shuffle(all_data)  
    male_s = all_data[:230]  
    female_s = all_data[230:]  
    means_difference.append(np.mean(male_s) - np.mean(female_s))
```

```
plt.figure(figsize=(6, 3))  
plt.hist(means_difference, edgecolor='k', linewidth=1.2, bins=12)  
plt.scatter(np.mean(male_spending) - np.mean(female_spending), 0,  
            color="r", s=40)  
plt.ylim(-10, 200)  
plt.show()
```



How surprising is an observed statistic?

- Out of the 1,000 permutations generated, how many were equal to or larger than the observed statistic?
 - We can count the number of simulated statistics that were equal to or larger than the observed statistic.
- The larger that number, the less “surprised” we are by the observed statistic.
- The smaller that number, the less likely that the spending of males and females are from the same distribution.

```
sum(np.array(means_difference) >= diff_of_means)
```

179

```
sum(np.abs(np.array(means_difference)) >= abs(diff_of_means))
```

335

one-sided

two-sided

The p-value

The p-value is a convention used to estimate how far out into the tails is considered “surprising.”

- Based on the area in the tail, either upper, lower, or both.
 - Since we are working on the empirical distribution, it's easy to compute.
- The p-value is the chance, based on the model in the **null hypothesis**, that the test statistic is equal to the value that was observed in the data or is even further in the direction of the alternative.

The p-value (cont'd.)



- If a p-value is small, then the observed statistic is far away from what the null hypothesis predicts.
 - The null hypothesis is unlikely, and the data supports the alternative.
 - Typically, if p-value is less than 0.05, it's considered “statistically significant.”
 - Reasonable to consider that occurrence in a frequency of 1/20 to be surprising.
- The **significance level** (α) is the probability of rejecting the null hypothesis when it is true. (Type-I error rate)
 - A significance level of 0.05 indicates a 5% risk of concluding that a difference exists when there is no actual difference.

General strategy for hypothesis testing

- 1. Calculate the observed test statistic:** Compute the summary metric (e.g., difference in means) from your actual sample data.
 - Identifying the appropriate statistic is critical for capturing the effect.
 - This is the **observed statistic** under the null hypothesis.
- 2. Construct the null distribution:** Use permutations or bootstrapping to simulate the test statistic's behavior **assuming the null hypothesis is true**.
 - This creates a distribution of outcomes that could occur purely by chance.
- 3. Evaluate plausibility (Calculate p-value):** Compare the **observed statistic** to the **null distribution** to see how “rare” the result is.
 - The **p-value**: How likely are we to observe a value as extreme as (or more extreme than) the observed statistic if the null hypothesis were true?